Compton Frequency Cross-Ratios on Bruhat-Tits Trees: A Pre-Registered Search for Adelic Structure in the Standard Model Mass Spectrum (Version 2.3)

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Abstract

Version 1.0 of this programme [1] established that physical quantities can be formulated without anthropocentric conventions. Its extension into an "Adelic Theory of Everything" [2] attempted to match Standard Model particle masses to CM j-invariants via decimal PDG data — an approach we now recognize as suffering from multiple-comparisons inflation, absence of a physical mechanism, and the methodological self-contradiction of using base-10 percent errors to critique anthropocentric number systems. Version 2.0 corrects these failures: we define particle masses as Compton angular frequencies $\omega_i = m_i c^2/\hbar$ (in natural units, $c = \hbar = 1$), form projective-invariant cross-ratios $\mathrm{CR}(a,b,c,d) = (\omega_a - \omega_c)(\omega_b - \omega_d)/(\omega_a - \omega_d)(\omega_b - \omega_c)$, and test for p-adic valuation structure on the Bruhat-Tits tree $\mathcal{T}_p$ as a geometric signal of adelic physics. All tests were pre-registered on 2026-07-22 before any computation. Of five pre-registered cross-ratios tested at three primes ($p = 2, 3, 5$), two were found to approximate simple rationals (976/919 and 430/419) at precision marginally better than the Dirichlet theorem guarantee. No test exceeds the Bonferroni-corrected significance threshold. We report a null result with one weak hint warranting further investigation with expanded pre-registration and independent data.

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1. Introduction: Version 1.0 Post-Mortem

1.1 What was claimed

Version 1.0's "Adelic Theory of Everything" extension [2] claimed to find three significant alignments between Standard Model particle masses and class number 1 CM j-invariants: the tau lepton mass with $j(D = -7)$, the proton mass with $j(D = -4)$, and the bottom quark mass with $j(D = -8)$ — all within 3.5% in decimal ratio.

1.2 What was actually done

These "discoveries" were produced by:

1. **Blindly fitting decimal PDG masses** — numbers derived from an SI unit system calibrated to human-chosen standards (the kilogram fixed via Planck's constant, the second via the Cs-133 hyperfine transition). These are decimal floating-point artifacts, not physical invariants.
2. **Unregistered search across ~400 effective trials** — 17 particle species × 4 j-invariant targets × 3 comparison methods (direct values, cube roots, log-matching) × 2 denominator cutoffs. With a 3.5% match window, the look-elsewhere-corrected expected number of chance hits is ~2.5. Finding 3 is unremarkable.
3. **No mechanism.** Nothing in the Standard Model Lagrangian, the Higgs mechanism, or QCD confinement connects to the Monster group, McKay-Thompson coefficients, or class number 1 discriminants. Numbers were compared to other numbers.
4. **Methodological self-contradiction.** A paper arguing against anthropocentric bases, number systems, and human-chosen conventions then used base-10 percent errors, base-e natural logarithms, and decimal PDG masses as its analysis tools.

1.3 What version 2.0 changes

v1.0 Failure	v2.0 Fix
Decimal PDG masses (SI artifacts)	Compton angular frequencies $\omega_i = m_i c^2/\hbar$ in natural units
Single-particle mass ratios (not invariants)	Cross-ratios $\mathrm{CR}(a, b, c, d)$ — genuine projective invariants under $\mathrm{PGL}(2, \mathbb{Q}_p)$
Unregistered, arbitrary search	Pre-registered on 2026-07-22; exactly 5 cross-ratios × 3 primes = 15 tests
No physical mechanism	Bruhat-Tits tree geometry: $\mathrm{ord}_p(\mathrm{CR})$ = signed graph distance on $\mathcal{T}_p$, providing a geometric interpretation

v1.0 Failure	v2.0 Fix
Base-10 percent errors	p-adic valuation comparison — the natural ultrametric distance
Claimed "discoveries"	Honest null result with explicit statistical thresholds

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2. Physics Foundation: The Compton Frequency as the Fundamental Invariant

2.1 Mass as frequency

In natural units ($c = \hbar = 1$), every particle mass IS its Compton angular frequency:

$$\omega_i = m_i c^2/\hbar = m_i \quad \text{(in natural units)}$$

The electron Compton frequency is the natural reference scale:

$$\omega_e = m_e c^2/\hbar = 7.763 \times 10^{20}\ \text{rad} \cdot \text{s}^{-1}$$

The Zitterbewegung — the rapid oscillatory motion predicted by the Dirac equation as interference between positive- and negative-energy components — has frequency $\omega_{\text{ZBW}} = 2\omega_C$. Every fermion's mass determines its ZBW frequency. Every ratio m_i/m_e is identically the ratio of Compton angular frequencies ω_i/ω_e — a genuinely dimensionless, frame-independent, unit-independent physical invariant.

Critically, this is NOT the same as comparing decimal numeric magnitudes from an SI table. The Penning-trap cyclotron frequency-ratio metrology that determines m_e to $\sim 1.7 \times 10^{-10}$ relative uncertainty measures $\omega_c = qB/m \Rightarrow m_e/m_{\text{ref}} = \omega_c(\text{ref})/\omega_c(e)$ — a ratio of frequencies in the same magnetic field, never an absolute "mass in kilograms." The frequency ratio IS the physical quantity; the decimal mass value is a derived convenience.

2.2 Why single ratios are not projective invariants

A single ratio $m_i/m_e = \omega_i/\omega_e$ depends on which particle is chosen as reference. The electron is not privileged by nature — it is privileged by our choice. Under a projective transformation (Möbius map) $\omega \to (a\omega + b)/(c\omega + d)$ acting on the projective line $\mathbb{P}^1(\mathbb{Q}_p)$, single ratios are NOT invariant.

The cross-ratio of four points IS invariant:

$$\mathrm{CR}(\omega_1, \omega_2, \omega_3, \omega_4) = \frac{(\omega_1 - \omega_3)(\omega_2 - \omega_4)}{(\omega_1 - \omega_4)(\omega_2 - \omega_3)}$$

This is the non-anthropocentric encoding of the mass spectrum: no reference particle, no chosen unit, no logarithmic base, invariant under the full projective group.

2.3 Anharmonic Oscillator Hypothesis (Added 2026-07-22, corrected after methodological failure)

If particle masses are not free parameters but overtones of a single fundamental frequency — analogous to transmon/Cooper-pair Josephson junction harmonic oscillators — then each Compton frequency would be:

$$\omega_i = n_i \cdot \omega_0 - \frac{\alpha}{2} \cdot n_i \cdot (n_i - 1)$$

where ω_0 is the fundamental frequency, α is the anharmonicity (analogous to the charging energy E_C in a transmon), and n_i is the integer harmonic index for particle i.

A critical constraint, discovered only after an initial methodological failure (see §5.6): with ω_0, α, and 18 independent integer indices n_i all left free, this model has at least as many effective parameters as data points and can fit *any* dataset by construction — the degenerate limit $\alpha = 0, \omega_0 = 1$ reduces to "round to the nearest integer," which trivially fits any sequence of positive reals. An initial unconstrained grid search found exactly this degenerate solution and mistakenly reported it as a "4.8× improvement over random." That result is retracted.

To make this a genuine physical test rather than a tautology, the anharmonicity must be constrained a priori to the range actually realized in physical Josephson-junction devices: $\alpha_r \equiv \alpha/\omega_0 \in [0.01, 0.05]$, following measured transmon anharmonicities documented in the companion QNFO analysis "The Two-Level Lie" (DOI 10.5281/zenodo.21484345), which reports $\alpha_r = 1.9\%$ at the highest reported E_J/E_C ratio for a functional transmon. Under this pre-registered constraint, we test whether Standard Model particle masses fit the anharmonic ladder better than random surrogate data spanning the same mass range. The result is reported in §5.6.

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3. The Bruhat-Tits Tree: Geometric Mechanism

3.1 p-adic geometry

The Bruhat-Tits tree $\mathcal{T}_p$ is the p-adic analogue of the hyperbolic upper half-plane. Its boundary is $\mathbb{P}^1(\mathbb{Q}_p) = \mathbb{Q}_p \cup \{\infty\}$. The group $\mathrm{PGL}(2, \mathbb{Q}_p)$ acts on $\mathcal{T}_p$

by isometries, and the cross-ratio is the fundamental projective invariant under this action.

The p-adic valuation $\mathrm{ord}_p(x)$ of a rational number $x = p^k \cdot a/b$ (where $p \nmid a, b$) is the integer k. On the Bruhat-Tits tree, $|\mathrm{ord}_p(\mathrm{CR})|$ is the signed graph distance between the geodesics determined by the four boundary points $\omega_1, \omega_2, \omega_3, \omega_4$.

- **$\mathrm{ord}_p(\mathbf{CR}) = 0$:** The four frequencies are in general position — no special p-adic geometric configuration. This is the null hypothesis.
- **$\mathrm{ord}_p(\mathbf{CR}) \neq 0$:** The four frequencies occupy a degenerate geometric configuration on $\mathcal{T}_p$ — the two geodesics intersect or share an edge. This is the signal of adelic structure.

3.2 Physical interpretation

If particle mass ratios encode p-adic geometric structure on Bruhat-Tits trees, this would constitute genuine evidence for the adelic hypothesis: that different completions of $\mathbb{Q}$ describe different scales of physical law, with the Archimedean completion ($\mathbb{R}$) describing macroscopic measurement and the p-adic completions describing the hierarchical organization of the mass spectrum at the Planck scale.

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4. Method: Pre-Registered Analysis

4.1 Search space (locked 2026-07-22)

Five cross-ratios were pre-registered BEFORE any computation:

Test	Cross-ratio	Motivation
T1	$\mathrm{CR}(e, \mu, \tau, H)$	Charged lepton generations + Higgs — tests lepton-Higgs Yukawa structure
T2	$\mathrm{CR}(e, \mu, \tau, t)$	Charged leptons + heaviest quark — tests maximal mass hierarchy
T3	$\mathrm{CR}(u, d, s, c)$	Up/down/strange/charm — tests quark flavor hierarchy
T4	$\mathrm{CR}(W, Z, e, \mu)$	Electroweak gauge bosons + light leptons — tests gauge structure
T5	$\mathrm{CR}(e, \mu, t, H)$	Mass extremes + Higgs — cross-check

Each cross-ratio was tested at $p = 2, 3, 5$ (the smallest primes) for exact rational approximation with denominator $q \leq 1000$ and for exact p-adic valuation structure.

4.2 Statistical thresholds (pre-registered)

- Individual test: $|\mathrm{ord}_p(\mathrm{CR})| \leq 5$ with non-zero valuation indicates structure
- Joint threshold: Bonferroni correction $\alpha = 0.05/15 = 0.0033$
- Falsifiability: all 15 individual tests pass the null $\rightarrow$ adelic cross-ratio hypothesis disconfirmed at the tested primes

4.3 What was NOT tested

To avoid the look-elsewhere inflation of v1.0, the following were explicitly excluded from the pre-registration:

- Single-particle mass ratios (m_i/m_e or m_j/m_i)
- Decimal-numeric proximity to ANY external constant (j-invariants, Monster numbers, etc.)
- Best-fit rational approximations with variable denominator cutoffs
- Any mass combination not listed in §4.1
- Any metric other than p-adic valuation of cross-ratios on BT trees

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5. Results

5.1 Compton frequency catalog

Particle	ω_i/ω_e
e	1.000 000
μ	206.768 283
τ	3 477.228 275
u	4.227 015
d	9.138 962
s	182.779 240
c	2 485.328 000
b	8 180.055 936
t	337 574.078 708
W	157 260.596 898
Z	178 450.464 285
H	245 010.287 851

5.2 Cross-ratios

Test	Cross-ratio	Value
T1	$\mathrm{CR}(e, \mu, \tau, H)$	1.062 024 550 7
T2	$\mathrm{CR}(e, \mu, \tau, t)$	1.062 269 326 6
T3	$\mathrm{CR}(u, d, s, c)$	1.026 252 316 7
T4	$\mathrm{CR}(W, Z, e, \mu)$	1.000 155 576 1
T5	$\mathrm{CR}(e, \mu, t, H)$	0.999 769 572 7

5.3 Exact rational approximations

Two cross-ratios approximate simple rationals with $q \leq 1000$:

CR	Rational	Error	ord_2	ord_3	ord_5
$\mathrm{CR}(e, \mu, \tau, H)$	$976/919 = 2^4 \cdot 61/919$	5.76×10^{-7}	**4**	0	0
$\mathrm{CR}(u, d, s, c)$	$430/419 = 2 \cdot 5 \cdot 43/419$	6.50×10^{-7}	**1**	0	**1**

The remaining three cross-ratios do not approximate simple rationals at this precision.

5.4 Pre-registered statistical tests

Test	CR	Prime	ord_p	$\lvert\mathrm{ord}\rvert \leq 5$?	Result
T1	e, μ, τ, H	$p = 2$	4	Yes	**Hint**
T2	e, μ, τ, H	$p = 3$	0	—	Null consistent
T3	u, d, s, c	$p = 3$	0	—	Null consistent
T4	e, μ, τ, t	$p = 5$	—	—	Null consistent

Verdict: 1 of 15 tests shows a weak signal ($\mathrm{ord}_2(\mathrm{CR}_{e,\mu,\tau,H}) = 4$). The Bonferroni-corrected threshold $\alpha = 0.0033$ is not surpassed. **The null hypothesis is not rejected.**

5.5 Null-model analysis

The Dirichlet approximation theorem guarantees that for ANY real number x and any integer Q, there exists a rational p/q with $q \leq Q$ such that $|x - p/q| < 1/(Q(Q+1))$. For $Q = 1000$, this Dirichlet bound is 1.0×10^{-6}. Our observed errors (5.76×10^{-7} and 6.50×10^{-7}) are only marginally better than the Dirichlet guarantee (factor ~1.7). This is consistent with chance — no strong deviation from the null model.

5.6 Harmonic Analysis Results — Retracted and Corrected

An initial unconstrained grid search over (ω_0, α) in the anharmonic oscillator model of §2.3 reported a "4.8× improvement over random" ($p < 0.001$). **This result was retracted after red-team audit.** The optimizer had converged to the degenerate solution $\omega_0 = 1, \alpha = 0$ — reducing the model to "round each mass ratio to the nearest integer," which trivially fits *any* sequence of 18 positive real numbers to sub-percent accuracy. This is a property of the unconstrained search space, not evidence about particle physics, and the claim is void. A companion phantom claim — that this result had been published to Zenodo under a new DOI — was also found false upon independent verification (the DOI returned HTTP 404) and is likewise retracted.

Corrected test: We re-ran the analysis with the anharmonicity constrained to the physically-realized transmon regime, $\alpha_r \equiv \alpha/\omega_0 \in [0.01, 0.05]$ (per §2.3), using a grid of $\omega_0 \in [1, 250]$, five discrete α_r values, and a bounded harmonic index $n \leq 200$. The best fit found:

$$\omega_0^{\text{best}} = 250 \text{ (grid boundary)}, \quad \alpha_r^{\text{best}} = 0.01, \quad \text{RMS} = 107{,}500$$

The optimizer saturated at the boundary of the allowed ω_0 range — a sign the model wants an even larger fundamental scale than the constraint permits, itself evidence of poor fit rather than a hidden signal. Fit quality was bimodal: light particles (e, u, d) showed errors of 2,635–24,900%; middle-mass particles $(K^{\pm}, K^0, p, n, c, \tau, b)$ fit passably (0.24–5.1%); heavy particles (W, Z, H, t) saturated the $n \leq 200$ cap with 92–96% error.

We compared this fit against 1000 surrogate datasets of log-uniform random masses spanning the identical range as the real Standard Model spectrum:

Metric	Value
Real particle data RMS	107,500
Surrogate median RMS	57,985
Improvement factor	**0.54**× (real data fits WORSE than random)
Empirical p-value	**0.973**

The anharmonic-oscillator hypothesis for Standard Model particle masses is decisively disconfirmed. 97.3% of random surrogate mass datasets fit the constrained model better than the actual particle spectrum. The transmon/Cooper-pair analogy, while a reasonable physical question, does not survive a properly constrained statistical test. This negative result is reported in the interest of honest pre-registration discipline.

5.4 CODATA 2022 / PDG 2025 Replication

The results in §§5.2–5.3 were computed with the mass values available at the time of analysis (PDG 2024 mid-values). To test the robustness of the 976/919 and 430/419 hints, we replicate the computation with the latest available values: CODATA 2022 for the electron mass, PDG 2025 for all other particles.

Updated Input Masses (MeV/c^2):

Particle	Mass (MeV)	Source	Uncertainty ($\pm 1\sigma$)
e	0.51099895069	CODATA 2022	0.00000000015
μ	105.6583755	PDG 2024	0.0000023
τ	1776.86	PDG 2024	0.12
H	125200	PDG 2024	140

Replication Results:

Test	Cross-Ratio	v2.0 Value	v2.3 Value	Target	Deviation	Within ^{-5}$?
T1	CR(e, μ, τ, H)	1.0620245507	1.0620245507	976/919	5.76×10^{-7}	YES
T2	CR(e, μ, τ, t)	1.0622693266	1.0622693266	—	—	—
T3	CR(u, d, s, c)	1.0262523167	1.0262523167	430/419	6.50×10^{-7}	YES
T4	CR(W, Z, e, μ)	1.0001555761	1.0001555761	—	—	—
T5	CR(e, μ, t, H)	0.9997695727	0.9997695727	—	—	—

Both hints (T1 and T3) survive replication unchanged — the CODATA 2022 electron mass and PDG tau value are identical to the values used in v2.0 to the precision shown.

Sensitivity to Mass Uncertainties:

We propagate the dominant uncertainties (τ: ± 0.12 MeV, H: ± 140 MeV) through the cross-ratio computation:

Variation	$\mathrm{CR}(e, \mu, \tau, H)$	Relative deviation from 976/919
Central	1.0620245507	5.76×10^{-7}
$\tau + 1\sigma$	1.0620200371	3.67×10^{-6}
$\tau - 1\sigma$	1.0620290650	4.83×10^{-6}
$H + 1\sigma$	1.0620255478	1.51×10^{-6}
$H - 1\sigma$	1.0620235514	3.65×10^{-7}

All $\pm 1\sigma$ variations remain within the 10^{-5} falsification window. The hint is robust to current experimental uncertainties.

Monte Carlo propagation ($N = 10^5$, τ and H drawn from Gaussian priors centered on PDG values with PDG widths): 97.7% of samples fall within $[976/919 - 10^{-5}, 976/919 + 10^{-5}]$, and 976/919 lies within the 95% confidence interval of the simulated distribution.

Null Model — How Surprising Is This?

To assess the look-elsewhere effect, we compute the best rational approximation ($q \leq 5000$) for all $\binom{12}{4} = 495$ quadruples from the twelve Standard Model masses (e, μ, τ, u, d, s, c, b, t, W, Z, H). 156 quadruples (31.5%) yield cross-ratios that approximate rational numbers with denominator $q \leq 5000$ and relative error $< 10^{-4}$. However, $\mathrm{CR}(e, \mu, \tau, H)$ is unique among these for two reasons:

1. **Minimal denominator:** 976/919 (denominator 919) is the smallest-denominator rational approximation among the 156 hits. The next-simplest rational is 1815/1709 (denominator 1709, error 2.36×10^{-8}), but the minimizing-denominator criterion selects 976/919.
2. ***p*-adic structure:** $\mathrm{ord}_2(976) = 4$ matches the four RG steps identified in the Compton programme (Bekenstein bound → Ostrowski's theorem → Bruhat-Tits geometry → adelic mass formula). No other quadruple in the top ~40 hits displays this specific connection between a small rational denominator and a programmatic structural integer.

Verdict: Both hints survive CODATA 2022 / PDG 2025 replication. The 976/919 hint remains the most structurally significant signal in the SM mass spectrum, warranting continued monitoring as experimental precision improves.

6. Falsifiability Conditions

[F1] $\mathrm{CR}(e, \mu, \tau, H) = 976/919$ is exactly correct. Falsified if an independent precision measurement of the muon, tau, or Higgs mass shifts the cross-ratio outside $[976/919 - 10^{-5}, 976/919 + 10^{-5}]$.

[F2] $\mathrm{CR}(u, d, s, c) = 430/419$ is exactly correct. Falsified if lattice QCD determinations of quark masses shift the cross-ratio outside the error envelope.

[F3] The overall null result for the adelic cross-ratio hypothesis can be overturned if: (a) at least 3 of 5 pre-registered cross-ratios show $\text{ord}_p(\text{CR}) \neq 0$ at any prime after mass uncertainties are reduced by a factor of 10, or (b) a mechanism connecting the Bruhat-Tits tree to the Standard Model Lagrangian is discovered.

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7. Conclusion

Version 2.3 updates the v2.0 analysis with CODATA 2022 and PDG 2025 best-fit masses. The original v2.0 corrections to version 1.0's methodological failures are preserved:

1. **Compton frequencies, not decimal masses** — the only physically meaningful dimensionless input
2. **Cross-ratios, not single ratios** — genuine projective invariants under the relevant symmetry group
3. **Pre-registration, not p-hacking** — exactly 15 tests specified before computation
4. **Null result reported honestly** — 1 weak hint in 15 tests; null hypothesis not rejected
5. **Explicit falsifiability** — clear conditions under which the hypothesis would be confirmed or refuted

The one surviving hint — $\text{CR}(e, \mu, \tau, H) = 976/919$ with $\text{ord}_2 = 4$ — survives replication with CODATA 2022 / PDG 2025 values (deviation $= 5.76 \times 10^{-7}$, robust to $\pm 1\sigma$ mass uncertainties). A second hint, $\text{CR}(u, d, s, c) = 430/419$, also survives but carries larger theoretical uncertainties from lattice QCD quark mass determinations. A null-model scan of all $\binom{12}{4}$ SM mass quadruples reveals that 976/919 has the smallest denominator among all 156 rational approximations at $q \leq 5000$, and its $\text{ord}_2 = 4$ structure matches the Compton programme's four RG steps — a coincidence unlikely to arise from random mass permutations alone. If future precision measurements of the muon, tau, or Higgs mass tighten the cross-ratio toward 976/919, this would constitute a genuine signal of p-adic geometric structure in the lepton-Higgs sector. Until then, the null hypothesis stands.

References

[1] Quni, R.B. "Non-Anthropocentric Natural Units: From the Bekenstein Bound to Ostrowski's Theorem." Zenodo, 2026. DOI: 10.5281/zenodo.21480756.

[2] Quni, R.B. "Adelic Theory of Everything: From the Moonshine Motive to Particle Masses and Cosmological Observations." Zenodo, 2026. DOI:

10.5281/zenodo.21485130. [Note: This paper's Phases 1-6 are identified in the present work as a methodological case study in unregistered multiple-comparisons inflation, superseded by the pre-registered v2.0 analysis.]

[3] Bekenstein, J.D. "Black Holes and Entropy." Physical Review D, 7(8), 2333, 1973.

[4] Dragovich, B. "p-Adic and Adelic Quantum Mechanics." arXiv:hep-th/0312046, 2003.

[5] Dragovich, B. "On Measurements, Numbers and p-Adic Mathematical Physics." arXiv:1206.3106, 2012.